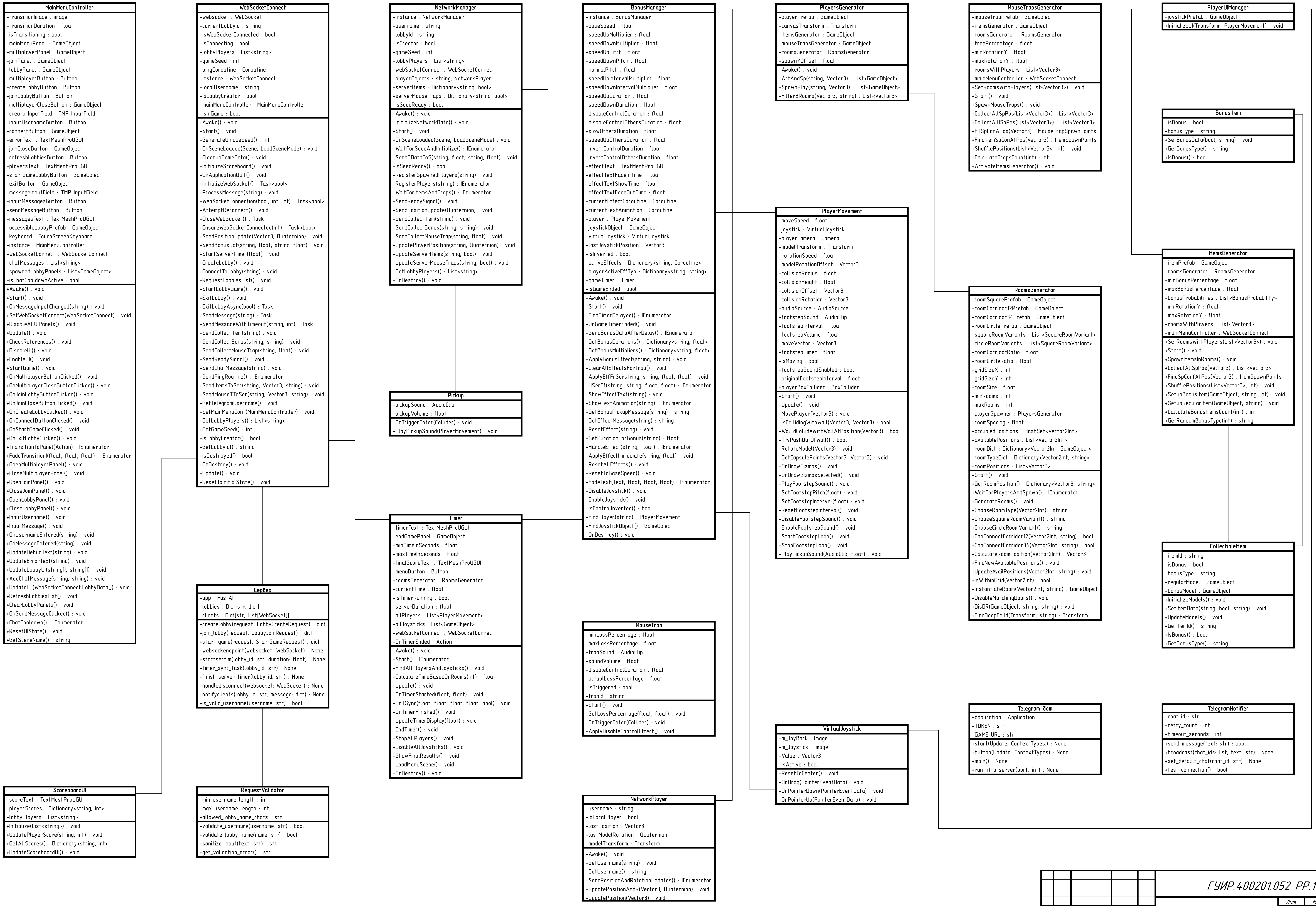
ГУИР.400201.052 РР.1



MainMenuController
-transitionImage : image
-transitionDuration : float
-isTransitioning : bool
-mainMenuPanel : GameObject
-multiplayerPanel : GameObject
-joinPanel : GameObject
-lobbyPanel : GameObject
-multiplayerButton : Button
-createLobbyButton : Button
-joinLobbyButton : Button
-multiplayerCloseButton : GameObject
-creatorInputField : TMP_InputField
-inputUsernameButton : Button
-connectButton : GameObject
-errorText : TextMeshProUGUI
-joinCloseButton : GameObject
-refreshLobbiesButton : Button
-playersText : TextMeshProUGUI
-startGameLobbyButton : GameObject
-exitButton : GameObject
-messageInputField : TMP_InputField
-inputMessagesButton : Button
-sendMessageButton : Button
-messagesText : TextMeshProUGUI
-accessibleLobbyPrefab : GameObject
-keyboard : TouchScreenKeyboard
-instance : MainMenuCpntroller
-webSocketConnect : WebSocketConnect
-chatMessages : List<string>
-spawnedLobbyPanels : List<GameObject>
-isChatCooldownActive : bool
+Awake() : void
+Start() : void
+OnMessageInputChanged(string) : void
+SetWebSocketConnect(WebSocketConnect) : void
+DisableAllUIPanels() : void
+Update() : void
+CheckReferences() : void
+DisableUI() : void
+EnableUI() : void
+StartGame() : void
+OnMultiplayerButtonClicked() : void
+OnMultiplayerCloseButtonClicked() : void
+OnJoinLobbyButtonClicked() : void
+OnJoinCloseButtonClicked() : void
+OnCreateLobbyClicked() : void
+OnConnectButtonClicked() : void
+OnStartGameClicked() : void
+OnExitLobbyClicked() : void
+TransitionToPanel(Action) : IEnumerator
+FadeTransitionI(float, float, float) : IEnumerator
+OpenMultiplayerPanel() : void
+CloseMultiplayerPanel() : void
+OpenJoinPanel() : void
+CloseJoinPanel() : void
+OpenLobbyPanel() : void
+CloseLobbyPanel() : void
+InputUsername() : void
+InputMessage() : void
+OnUsernameEntered(string) : void
+OnMessageEntered(string) : void
+UpdateDebugText(string) : void
+UpdateErrorText(string) : void
+UpdateLobbyUI(string[], string[]) : void
+AddChatMessage(string, string) : void
+UpdateLL(WebSocketConnect.LobbyData[]) : void
+RefreshLobbiesList() : void
+ClearLobbyPanels() : void
+OnSendMessageClicked() : void
+ChatCooldown() : IEnumerator
+ResetUIState() : void
+GetSceneName() : string

WebSocketConnect
-websocket : WebSocket
-currentLobbyId : string
-isWebSocketConnected : bool
-isConnecting : bool
-lobbyPlayers : List<string>
-gameSeed : int
-pingCoroutine : Coroutine
-instance : WebSocketConnect
-localUsername : string
-isLobbyCreator : bool
-mainMenuController : MainMenuController
-isInGame : bool
+Awake() : void
+Start() : void
+GenerateUniqueSeed() : int
+OnSceneLoaded(Scene, LoadSceneMode) : void
+CleanupGameData() : void
+InitializeScoreboard() : void
+OnApplicationQuit() : void
+InitializeWebSocket() : Task<bool>
+ProcessMessage(string) : void
+WebSocketConnection(bool, int, int) : Task<bool>
+AttemptReconnect() : void
+CloseWebSocket() : Task
+EnsureWebSocketConnected(int) : Task<bool>
+SendPositionUpdate(Vector3, Quaternion) : void
+SendBonusDat(string, float, string, float) : void
+StartServerTimer(float) : void
+CreateLobby() : void
+ConnectToLobby(string) : void
+RequestLobbiesList() : void
+StartLobbyGame() : void
+ExitLobby() : void
+ExitLobbyAsync(bool) : Task
+SendMessage(string) : Task
+SendMessageWithTimeout(string, int) : Task
+SendCollectItem(string) : void
+SendCollectBonus(string, string) : void
+SendCollectMouseTrap(string, float) : void
+SendReadySignal() : void
+SendChatMessage(string) : void
+SendPingRoutine() : IEnumerator
+SendItemsToSer(string, Vector3, string) : void
+SendMouseTToSer(string, Vector3, string) : void
+GetTelegramUsername() : void
+SetMainMenuCont(MainMenuController) : void
+GetLobbyPlayers() : List<string>
+GetGameSeed() : int
+IsLobbyCreator() : bool
+GetLobbyId() : string
+IsDestroyed() : bool
+OnDestroy() : void
+Update() : void
+ResetToInitialState() : void

NetworkManager
-Instance : NetworkManager
-username : string
-lobbyId : string
-isCreator : bool
-gameSeed : int
-lobbyPlayers : List<string>
-webSocketConnect : WebSocketConnect
-playerObjects : string, NetworkPlayer
-serverItems : Dictionary<string, bool>
-serverMouseTraps : Dictionary<string, bool>
-isSeedReady : bool
+Awake() : void
+InitializeNetworkData() : void
+Start() : void
+OnSceneLoaded(Scene, LoadSceneMode) : void
+WaitForSeedAndInitialize() : IEnumerator
+SendBDataToS(string, float, string, float) : void
+IsSeedReady() : bool
+RegisterSpawnedPlayers(string) : void
+RegisterPlayers(string) : IEnumerator
+WaitForItemsAndTraps() : IEnumerator
+SendReadySignal() : void
+SendPositionUpdate(Quaternion) : void
+SendCollectItem(string) : void
+SendCollectBonus(string, string) : void
+SendCollectMouseTrap(string, float) : void
+UpdatePlayerPosition(string, Quaternion) : void
+UpdateServerItems(string, bool) : void
+UpdateServerMouseTraps(string, bool) : void
+GetLobbyPlayers() : List<string>
+OnDestroy() : void

BonusManager
-Instance : BonusManager
-baseSpeed : float
-speedUpMultiplier : float
-speedDownMultiplier : float
-speedUpPitch : float
-speedDownPitch : float
-normalPitch : float
-speedUpIntervalMultiplier : float
-speedDownIntervalMultiplier : float
-speedUpDuration : float
-speedDownDuration : float
-disableControlDuration : float
-disableControlOthersDuration : float
-slowOthersDuration : float
-speedUpOthersDuration : float
-invertControlDuration : float
-invertControlOthersDuration : float
-effectText : TextMeshProUGUI
-effectTextFadeInTime : float
-effectTextShowTime : float
-effectTextFadeOutTime : float
-currentEffectCoroutine : Coroutine
-currentTextAnimation : Coroutine
-player : PlayerMovement
-joystickObject : GameObject
-virtualJoystick : VirtualJoystick
-lastJoystickPosition : Vector3
-isInverted : bool
-activeEffects : Dictionary<string, Coroutine>
-playerActiveEffTyp : Dictionary<string, string>
-gameTimer : Timer
-isGameEnded : bool
+Awake() : void
+Start() : void
+FindTimerDelayed() : IEnumerator
+OnGameTimerEnded() : void
+SendBonusDataAfterDelay() : IEnumerator
+GetBonusDurations() : Dictionary<string, float>
+GetBonusMultipliers() : Dictionary<string, float>
+ApplyBonusEffect(string, string) : void
+ClearAllEffectsForTrap() : void
+ApplyEffFrSerstring, string, float, float) : void
+HSerEf(string, string, float, float) : IEnumerator
+ShowEffectText(string) : void
+ShowTextAnimation(string) : IEnumerator
+GetBonusPickupMessage(string) : string
+GetEffectMessage(string) : string
+ResetEffect(string) : void
+GetDurationForBonus(string) : float
+HandleEffect(string, float) : IEnumerator
+ApplyEffectImmediate(string, float) : void
+ResetAllEffects() : void
+ResetToBaseSpeed() : void
+FadeText(Text, float, float, float) : IEnumerator
+DisableJoystick() : void
+EnableJoystick() : void
+IsControlInverted() : bool
+FindPlayer(string) : PlayerMovement
+FindJoystickObject() : GameObject
+OnDestroy() : void

PlayersGenerator
-playerPrefab : GameObject
-canvasTransform : Transform
-itemsGenerator : GameObject
-mouseTrapsGenerator : GameObject
-roomsGenerator : RoomsGenerator
-spawnYOffset : float
+Awake() : void
+ActAndSp(string, Vector3) : List<GameObject>
+SpawnPlay(string, Vector3) : List<GameObject>
+FilterBRooms(Vector3, string) : List<Vector3>

MouseTrapsGenerator
-mouseTrapPrefab : GameObject
-itemsGenerator : GameObject
-roomsGenerator : RoomsGenerator
-trapPercentage : float
-minRotationY : float
-maxRotationY : float
-roomsWithPlayers : List<Vector3>
-mainMenuController : WebSocketConnect
+SetRoomsWithPlayers(List<Vector3>) : void
+Start() : void
+SpawnMouseTraps() : void
+CollectAllSpPos(List<Vector3>) : List<Vector3>
+CollectAllISpPos(List<Vector3>) : List<Vector3>
+FTSpConAPos(Vector3) : MouseTrapSpawnPoints
+FindItemSpConAtPos(Vector3) : ItemSpawnPoints
+ShufflePositions(List<Vector3>, int) : void
+CalculateTrapsCount(int) : int
+ActivateItemsGenerator() : void

PlayerUIManager
-joystickPrefab : GameObject
+InitializeUI(Transform, PlayerMovement) : void

BonusItem
-isBonus : bool
-bonusType : string
+SetBonusData(bool, string) : void
+GetBonusType() : string
+IsBonus() : bool

PlayerMovement
-moveSpeed : float
-joystick : VirtualJoystick
-playerCamera : Camera
-modelTransform : Transform
-rotationSpeed : float
-modelRotationOffset : Vector3
-collisionRadius : float
-collisionHeight : float
-collisionOffset : Vector3
-collisionRotation : Vector3
-audioSource : AudioSource
-footstepSound : AudioClip
-footstepInterval : float
-footstepVolume : float
-moveVector : Vector3
-footstepTimer : float
-isMoving : bool
-footstepSoundEnabled : bool
-originalFootstepInterval : float
-playerBoxCollider : BoxCollider
+Start() : void
+Update() : void
+MovePlayer(Vector3) : void
+IsCollidingWithWall(Vector3, Vector3) : bool
+WouldCollideWithWallAtPosition(Vector3) : bool
+TryPushOutOfWall() : bool
+RotateModel(Vector3) : void
+GetCapsulePoints(Vector3, Vector3) : void
+OnDrawGizmos() : void
+OnDrawGizmosSelected() : void
+PlayFootstepSound() : void
+SetFootstepPitch(float) : void
+SetFootstepInterval(float) : void
+ResetFootstepInterval() : void
+DisableFootstepSound() : void
+EnableFootstepSound() : void
+StartFootstepLoop() : void
+StopFootstepLoop() : void
+PlayPickupSound(AudioClip, float) : void

ItemsGenerator
-itemPrefab : GameObject
-roomsGenerator : RoomsGenerator
-minBonusPercentage : float
-maxBonusPercentage : float
-bonusProbabilities : List<BonusProbability>
-minRotationY : float
-maxRotationY : float
-roomsWithPlayers : List<Vector3>
-mainMenuController : WebSocketConnect
+SetRoomsWithPlayers(List<Vector3>) : void
+Start() : void
+SpawnItemsInRooms() : void
+CollectAllSpPos(Vector3) : List<Vector3>
+FindSpConfAtPos(Vector3) : ItemSpawnPoints
+ShufflePositions(List<Vector3>, int) : void
+SetupBonusItem(GameObject, string, int) : void
+SetupRegularItem(GameObject, string) : void
+CalculateBonusItemsCount(int) : int
+GetRandomBonusType(int) : string

RoomsGenerator
-roomSquarePrefab : GameObject
-roomCorridor12Prefab : GameObject
-roomCorridor34Prefab : GameObject
-roomCirclePrefab : GameObject
-squareRoomVariants : List<SquareRoomVariant>
-circleRoomVariants : List<SquareRoomVariant>
-roomCorridorRatio : float
-roomCircleRatio : float
-gridSizeX : int
-gridSizeY : int
-roomSize : float
-minRooms : int
-maxRooms : int
-playerSpawner : PlayersGenerator
-roomSpacing : float
-occupiedPositions : HashSet<Vector2Int>
-availablePositions : List<Vector2Int>
-roomDict : Dictionary<Vector2Int, GameObject>
-roomTypeDict : Dictionary<Vector2Int, string>
-roomPositions : List<Vector3>
+Start() : void
+GetRoomPosition() : Dictionary<Vector3, string>
+WaitForPlayersAndSpawn() : IEnumerator
+GenerateRooms() : void
+ChooseRoomType(Vector2Int) : string
+ChooseSquareRoomVariant() : string
+ChooseCircleRoomVariant() : string
+CanConnectCorridor12(Vector2Int, string) : bool
+CanConnectCorridor34(Vector2Int, string) : bool
+CalculateRoomPosition(Vector2Int) : Vector3
+FindNewAvailablePositions() : void
+UpdateAvailPositions(Vector2Int, string) : void
+IsWithinGrid(Vector2Int) : bool
+InstantiateRoom(Vector2Int, string) : GameObject
+DisableMatchingDoors() : void
+DisDR(GameObject, string, string) : void
+FindDeepChild(Transform, string) : Transform

Pickup
-pickupSound : AudioClip
-pickupVolume : float
+OnTriggerEnter(Collider) : void
+PlayPickupSound(PlayerMovement) : void

Timer
-timerText : TextMeshProUGUI
-endGamePanel : GameObject
-minTimeInSeconds : float
-maxTimeInSeconds : float
-finalScoreText : TextMeshProUGUI
-menuButton : Button
-roomsGenerator : RoomsGenerator
-currentTime : float
-isTimerRunning : bool
-serverDuration : float
-allPlayers : List<PlayerMovement>
-allJoysticks : List<GameObject>
-webSocketConnect : WebSocketConnect
-OnTimerEnded : Action
+Awake() : void
+Start() : IEnumerator
+FindAllPlayersAndJoysticks() : void
+CalculateTimeBasedOnRooms(int) : float
+Update() : void
+OnTimerStarted(float, float) : void
+OnTSync(float, float, float, float, bool) : void
+OnTimerFinished() : void
+UpdateTimerDisplay(float) : void
+EndTimer() : void
+StopAllPlayers() : void
+DisableAllJoysticks() : void
+ShowFinalResults() : void
+LoadMenuScene() : void
+OnDestroy() : void

CollectibleItem
-itemId : string
-isBonus : bool
-bonusType : string
-regularModel : GameObject
-bonusModel : GameObject
+InitializeModels() : void
+SetItemData(string, bool, string) : void
+UpdateModels() : void
+GetItemId() : string
+IsBonus() : bool
+GetBonusType() : string

Сервер
-app : FastAPI
-lobbies : Dict[str, dict]
-clients : Dict[str, List[WebSocket]]
+createlobby(request: LobbyCreateRequest) : dict
+join_lobby(request: LobbyJoinRequest) : dict
+start_game(request: StartGameRequest) : dict
+websockendpoint(websocket: WebSocket) : None
+startsertim(lobby_id: str, duration: float) : None
+timer_sync_task(lobby_id: str) : None
+finish_server_timer(lobby_id: str) : None
+handledisconnect(websocket: WebSocket) : None
+notifyclients(lobby_id: str, message: dict) : None
+is_valid_username(username: str) : bool

MouseTrap
-minLossPercentage : float
-maxLossPercentage : float
-trapSound : AudioClip
-soundVolume : float
-disableControlDuration : float
-actualLossPercentage : float
-isTriggered : bool
-trapId : string
+Start() : void
+SetLossPercentage(float, float) : void
+OnTriggerEnter(Collider) : void
+ApplyDisableControlEffect() : void

Telegram-бот
-application : Application
-TOKEN : str
-GAME_URL : str
+start(Update, ContextTypes.) : None
+button(Update, ContextTypes) : None
+main() : None
+run_http_server(port: int) : None

TelegramNotifier
-chat_id : str
-retry_count : int
-timeout_seconds : int
+send_message(text: str) : bool
+broadcast(chat_ids: list, text: str) : None
+set_default_chat(chat_id: str) : None
+test_connection() : bool

VirtualJoystick
-m_JoyBack : Image
-m_Joystick : Image
-Value : Vector3
-IsActive : bool
+ResetToCenter() : void
+OnDrag(PointerEventData) : void
+OnPointerDown(PointerEventData) : void
+OnPointerUp(PointerEventData) : void

ScoreboardUI
-scoreText : TextMeshProUGUI
-playerScores : Dictionary<string, int>
-lobbyPlayers : List<string>
+Initialize(List<string>) : void
+UpdatePlayerScore(string, int) : void
+GetAllScores() : Dictionary<string, int>
+UpdateScoreboardUI() : void

RequestValidator
-min_username_length : int
-max_username_length : int
-allowed_lobby_name_chars : str
+validate_username(username: str) : bool
+validate_lobby_name(name: str) : bool
+sanitize_input(text: str) : str
+get_validation_error() : str

ГУИР.400201.052 РР.1

Изм.	Лист	№ докум.	Подп.	Дата
Разраб.		Казаков		
Пров.		Смирнов		
Т. контр.		Смирнов		
Реценз.				
Н. контр.		Страковuч		
Утв.		Никульшин		

Многопользовательская 3D игра с интеграцией в Telegram.
Диаграмма классов

Лит.	Масса	Масштаб
Т		

Лист | Листов 1

ЭВМ, гр. 250501